

Pranav Dangi

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EDUCATION

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| • National University of Singapore (NUS) | Singapore |
| • PhD Student, School of Computing | 2024 - 202x |
| • Birla Institute of Technology and Science, Pilani (BITS Pilani) | Pilani, India |
| • B.E. (Hons.) Electrical and Electronics Engineering | 2019 - 2023 |
- Thesis: Addressing Sparsity through Algorithms, Memory and Compute

AWARDS, SCHOLARSHIPS AND ACHIEVEMENTS

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- ASPLOS 2026 Best Paper Honorable Mention
 - President's Graduate Fellowship (University's Highest Fellowship for Research and Academic Excellence), 2024-Now
 - NUS Research Achievement Award (Award for Outstanding Research Achievements), 2024, 2025
 - Travel Grant: MICRO 2025, ASPLOS 2025, 2026, DAC 2024
 - Google Summer of Code Fellowship, 2021
 - BITS Institute Dean's MCN Scholarship, 2019-20, 2021-23

PUBLICATIONS (*EQUAL CONTRIBUTION) (CHRONOLOGICAL ORDER)

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- **Pranav Dangi**, Rohan Juneja, Lieven Eeckhout, Tulika Mitra. "Leveraging CGRA Programmability for Reliable and Sustainable Hardware Acceleration", (*Under Review*)
 - Bai Zhenyu, Li Wei, Heru Wang, **Pranav Dangi**, Cheng Tan, Wong Weng Fai, Tulika Mitra. "TileLoom: Improving the Programmability of Spatial-Dataflow Architectures Using Tile-Based Languages with Automatic Dataflow Planning", *20th USENIX Conference on Operating Systems Design and Implementation (OSDI)*, 2026
 - Bai Zhenyu*, **Pranav Dangi***, Rohan Juneja, Zhaoying Li, Zhanglu Yan, Huiying Lan, Tulika Mitra. "A Data-Driven Dynamic Execution Orchestration Architecture", *31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2026
[Best Paper Honorable Mention]
 - Rohan Juneja, **Pranav Dangi**, Thilini Kaushalya Bandara, Tulika Mitra, Li-Shiuan Peh. "Nexus Machine: An Active Message Inspired Reconfigurable Architecture for Irregular Workloads", *58th IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2025
 - Rohan Juneja, **Pranav Dangi**, Thilini Kaushalya Bandara, Zhaoying Li, Dhananjaya Wijerathne, Li-Shiuan Peh, Tulika Mitra "Building an Open CGRA Ecosystem for Agile Innovation" *44th ACM/IEEE International Conference on Computer Aided Design, (ICCAD) 2025 (Invited Paper)*
 - Zhaoying Li*, **Pranav Dangi***, Chenyang Yin, Thilini Kaushalya Bandara, Rohan Juneja, Cheng Tan, Tulika Mitra. "Enhancing CGRA Efficiency through Aligned Compute and Communication Provisioning", *30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2025
 - **Pranav Dangi***, Thilini Kaushalya Bandara*, Saeideh Sheikhpour, Tulika Mitra, Lieven Eeckhout. "Sustainable Hardware Specialization", *43rd ACM/IEEE International Conference on Computer Aided Design (ICCAD)*, 2024
 - **Pranav Dangi**, Bai Zhenyu, Rohan Juneja, Dhananjaya Wijerathne, Tulika Mitra. "A Generalized Accelerator for Variably Sparse Matrix Computations in ML", *33rd International Conference on Parallel Architectures and Compilation Techniques (PACT)*, 2024
 - Bai Zhenyu, Wu Dan, **Pranav Dangi**, Dhananjaya Wijerathne, Pavan Miriyala, Tulika Mitra. "Irregular Data-aware dynamic scheduling of workloads on heterogeneous systems", *IEEE Transactions on Computers (TC)*, 2025
 - Bai Zhenyu, **Pranav Dangi**, Huize Li, Tulika Mitra. "Scalable, Efficient Window-Attention Based Transformer Acceleration", *61st ACM/IEEE Design Automation Conference (DAC)*, 2024

PAST RESEARCH EXPERIENCE

- **NUS School Of Computing** Singapore
Undergraduate Research Assistant 2023-2024
- **TIFR/TCS Research** Mumbai
Research Intern 2022

PAST ENGINEERING EXPERIENCE

- **Intel** Bangalore
Intern, SoC design 2022
- **Google Summer of Code** Remote
Fellow, RTEMS 2021

TEACHING

- **Graduate Teaching Assistant** Singapore
GEI1000 Computational Thinking 2024
- **Undergraduate Teaching Assistant** Pilani, India
CS F342 Computer Architecture; EEE F313 Analog, Digital VLSI Design 2022